Management Information Systems

Week 6

Kangah Park (k.park@postech.ac.kr)

POSTECH
POHANG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Last Week…

- Enterprise systems integrate the whole firm through connected software modules and a shared central database
- SCM systems improve SC decisions by automating information flows across partners, helping firms decide what to buy, make, and ship while reducing uncertainty and the ***bullwhip effect***.
- CRM systems create a company-wide view of customers by integrating ***sales, marketing, and service***
- Successful implementation is ***difficult and costly*** because enterprise applications require major organizational change, employee training, process redesign, etc.
- Enterprise applications are becoming more ***flexible*** through web-based integration, cloud and mobile deployment, and added social networking capabilities.

Weekly Topics

- **Subject to change without prior notice**

W	Ch	Topic	W	Ch	Topic
1		Orientation Introduction to MIS	9	5	IT infrastructure and emerging technologies
2	1	Information Systems in Global business today	10	6	Foundations of Business Intelligence
3	2	Global E-business and Collaboration	11		*No Class*
4	3	Information Systems, Organizations, and Strategy	12	11	Enhancing decision making
5	9	Achieving operational excellence and customer intimacy	13	12	Managing knowledge
6	**4**	**Ethical and Social issues in information systems**	14	13	Building information systems / Hands-on Session
7	7	Telecommunications, the internet, and wireless technology*	15	8	Securing information systems*
8		*Midterm Exam*	*16*		*Final Exam*

Chapter 4.

Ethical and Social Issues in Information Systems

Learning Objectives

1. **What ethical, social, and political *issues* are raised by information systems?**

2. **What specific *principles for conduct* can be used to guide ethical decisions?**

3. **Why do contemporary information systems technology and the Internet pose *challenges* to the protection of individual privacy and intellectual property?**

4. **How have information systems *affected* laws for establishing accountability, liability, and the quality of everyday life?**

4.1

What ethical, social, and political issues are raised by information systems?

The most ethically challenging periods

- **Recent cases of failed ethical judgment in business**
  - Volkswagen AG, Wells Fargo, General Motors, Takata Corporation
  - In many, information systems used to bury decisions from public scrutiny

Volswagen AG	Installed "defeat-device" emissions software on over 500,000 diesel cars in the United States and roughly 10.5 million more worldwide that allowed them to meet U.S. emissions standards during regulatory testing while actually spewing unlawful levels of pollutants into the air in real-world driving. Criminal charges were brought against six ranking VW executives, including Oliver Schmidt, who was sentenced to seven years in prison and a $400,000 fine.
Wells Fargo (2018)	Wells Fargo bank admitted to opening millions of false accounts, manipulating terms of mortgages, and forcing auto loan customers to purchase unneeded insurance. The bank was fined $2.5 billion by the federal government.
General Motors, Inc. (2015)	General Motors CEO admitted the firm covered up faulty ignition switches for more than a decade, resulting in the deaths of at least 114 customers. More than 100 million vehicles worldwide were affected.
Takata Corporation (2015)	Takata executives admitted they covered up faulty airbags used in millions of cars over many years. Three executives were indicted on criminal charges and Takata was fined $1 billion. Takata filed for bankruptcy in June 2017.

Deloitte Case

- **Deloitte delivered a government report containing AI-generated fake references and fabricated quotes**
- **The firm refunded part of its fee and faced public criticism for insufficient human oversight in its AI-assisted work.**

Search

FORTUNE

Deloitte was caught using AI in $290,000 report to help the Australian government crack down on welfare after a researcher flagged hallucinations

By Nino Paoli
News Fellow
October 7, 2025, 5:10 PM ET

Add us on

Source: https://fortune.com/2025/10/07/deloitte-ai-australia-government-report-hallucinations-technology-290000-refund/

Legal Environment and Role of Information Systems

- **Legal Environment and Consequences:**
  - The new legal environment imposes rigid ***penalties*** for managers including imprisonment convicted of violating laws
  - Federal sentencing guidelines mandate severe penalties for executives based on various factors

- **Role of Information Systems:**
  - Information systems were instrumental in many fraud cases, enabling perpetrators to hide their actions.

Ethics and Information Systems

- **Ethics:** principles of *right and wrong* that individuals use to make choices to guide their behaviors

- **Internet and digital technologies have intensified ethical concerns, such as customer data use, privacy protection, and intellectual property rights.**
  - Other pressing ethical issues include establishing accountability, ensuring system quality and safety, and preserving societal values.

- **It's crucial to consider the ethical and socially responsible course of action when using information systems.**

A Need for Rules, Norms, and Guidance

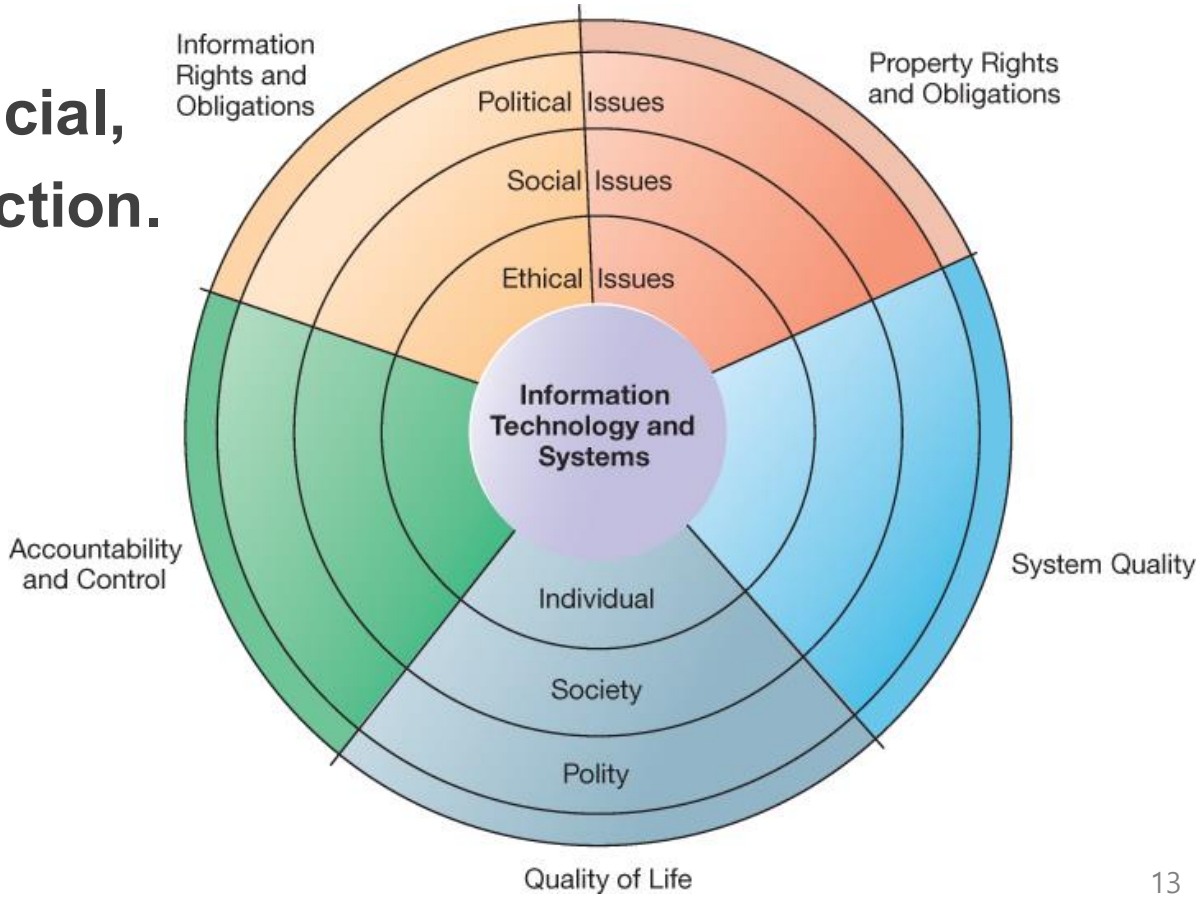
- **Ethical issues in information systems are closely connected to broader social and political debates.**
- **New information technology *disrupts existing norms*, creating situations that current rules do not fully cover.**
- **Laws and social institutions *take time* to adapt and develop new norms, responsibilities, and expectations.**
- **During this gap, individuals often must act in legal and ethical gray areas without clear guidance.**

Key Technology Trends that Raise Ethical Issues

- **Doubling of Computing Power:** Increased dependence on systems raises vulnerability to errors and data quality issues, yet standards for ensuring accuracy are lacking.
- **Advances in Data Storage:** Proliferation of databases allows inexpensive and effective violation of individual privacy
- **Advances in Data Analysis Techniques:** Companies and government agencies can obtain highly detailed personal information ➔ concerns about privacy and profiling.
- **Networking Advances:** Reduced costs of data movement and access, along with remote data mining capabilities, allow for unprecedented invasion of privacy.

A Model for Thinking about Ethical, Social, and Political Issues

- **The model helps illustrate the dynamics connecting ethical, social, and political issues.**
- **It facilitates the identification of *key moral dimensions* in the information society, spanning individual, social, and political levels of action.**



Five Moral Dimensions of the Information Age

1. **Information Rights and Obligations:** individuals' and organizations' rights regarding their own information and what they can protect.

2. **Property Rights and Obligations:** Addresses the protection of traditional intellectual property rights in a digital society

3. **Accountability and Control:** Examines who can be held accountable for harm to information and property rights and how control is managed

4. **System Quality:** Considers the standards of data and system quality necessary to safeguard individual rights and societal safety.

5. **Quality of Life:** Explores the preservation of values in an information-based society, protection of institutions, and support for cultural values by new technology.

4.2

What specific principles for conduct can be used to guide ethical decisions?

Basic Concepts

- **Responsibility:** Individuals making ethical choices accept the potential costs, duties, and obligations of their actions.

- **Accountability:** Systems and institutions must have mechanisms in place to determine who took action and who is responsible for it.

- **Liability:** Extends responsibility ***into the legal domain***, allowing individuals to recover damages through established laws.

- **Due Process:** Ensures that ***laws are known, understood, and applied correctly,*** providing a framework for resolving disputes with an ability to appeal to higher authorities

Ethical Analysis: A Five-Step Process

1. **Identify and Describe Facts:** Clearly outline the ***facts*** of the situation, ensuring accuracy and agreement among involved parties.

2. **Define the Conflict or Dilemma:** Identify ***conflicting values***, recognizing opposing courses of action that support different higher-order values

3. **Identify Stakeholders:** Recognize ***all parties with an interest*** in the outcome and understand their perspectives and desires.

4. **Identify Options:** Explore potential courses of ***action***, recognizing that some options may better satisfy interests than others.

5. **Consider Consequences:** Evaluate the potential ***consequences*** of each option, considering both ethical correctness and broader impacts over time.

Candidate Ethical Principles

1. ***Golden Rule*: Treat others as you would like to be treated.**
   - Putting yourself in the place of others
2. **Immanuel Kant's *Categorical Imperative*: Only take actions that could be universalized without contradiction.**
   - "If everyone did this, could the organization or society survive?"
3. ***Slippery slope rule*: Avoid actions that, if repeated, could lead to unacceptable consequences in the long run.**
   - Consider the ***potential long-term impacts*** of decisions.

Candidate Ethical Principles (Cont.)

4. ***Utilitarian Principle*: Take the action that achieves the higher or greater value**
5. ***Risk Aversion Principle*: Take the action that produces the least harm or potential cost**
6. **Ethical "*No Free Lunch*" Rule: Assume that virtually all tangible and intangible objects are owned by someone unless there is a specific declaration otherwise.**
   - Respect the property of others and be prepared to ***compensate*** for its use.

Image Source: Boston Solar

Professional Codes of Conduct

- **Professionals are granted special rights and obligations because of their expertise and the trust society places in them.**

- **Professional associations establish *codes of conduct* to guide and regulate members' behavior.**

- **These groups also set *qualification and competency standards* to control entry into the profession and maintain integrity.**

- **Codes of ethics outline *key principles* such as avoiding harm, respecting property rights, and protecting privacy**

Some Real-World Ethical Dilemmas

Image: Newman web solutions

Voice Recognition Software

- **Competing values:** company's desire to increase productivity and reduce costs vs. potential impact on employees losing their jobs.
- **Ethical Analysis:**
  - ***Golden Rule***: Consider how you would feel if you were ***in the employee's position***. Would you want your job replaced by technology?
  - ***Utilitarian Principle***: Does the benefit of increased productivity outweigh the harm of job loss?
- **Possible Solution:** Implement training programs or reassignment opportunities for affected employees to mitigate the negative impact of job loss while still benefiting from productivity gains.

Some Real-World Ethical Dilemmas

Employee Internet Monitoring

- **Competing Values:** Employer's right to protect company resources and maintain productivity vs. employees' right to privacy and autonomy
- **Ethical Analysis:**
  - ***Categorical Imperative***: Consider whether monitoring employees' internet activity would be acceptable if everyone did it. Would it infringe on employees' rights?
  - ***Risk Aversion Principle***: Evaluate the ***potential harm*** caused by excessive monitoring, such as erosion of trust and morale among employees.

- **Possible Solution:** Establish clear policies on internet usage that balance the company's need for productivity with respect for employees' privacy. Implement monitoring systems transparently and ensure they are used carefully.

Discussion

- **If my AI agent automatically executes instructions on my behalf and makes a mistake, who should be held responsible: the user who gave the instruction, the company deploying the AI, or the developer of the AI system?**

Image Source: OpenClaw.ai

4.3

Why do contemporary information systems technology and the Internet pose challenges to the protection of individual privacy and intellectual property?

Privacy and Freedom in the Internet Age

- ***Privacy*** **refers to an individual's right to be left alone, free from surveillance or interference by others, and to control information about oneself.**

- **Advances in IT have made *invasions of privacy* cheaper, more profitable, and more effective, especially through digital and high-tech surveillance in the workplace.**

- **In response, many countries, including the United States, Canada, and Germany, have established constitutional and legal protections to safeguard privacy rights.**

European Directive on Data Protection

- **EU General Data Protection Regulation (GDPR):**
  - Implemented in 2018, it's a significant privacy legislation, applying globally to firms handling EU citizens' personal data.
  - Replaces the 1998 Data Protection Directive, offering ***stronger privacy protection than U.S. laws.***
- **GDPR Key Objectives:**
  - Strengthen individuals' rights to personal data
  - Harmonize data protection standards across EU nations
  - Enforce regulations globally for organizations dealing with EU citizen data.
- **Individual Rights Under GDPR:**
  - Access to personal data, deletion (right to be forgotten), data portability, and right to sue for damages.

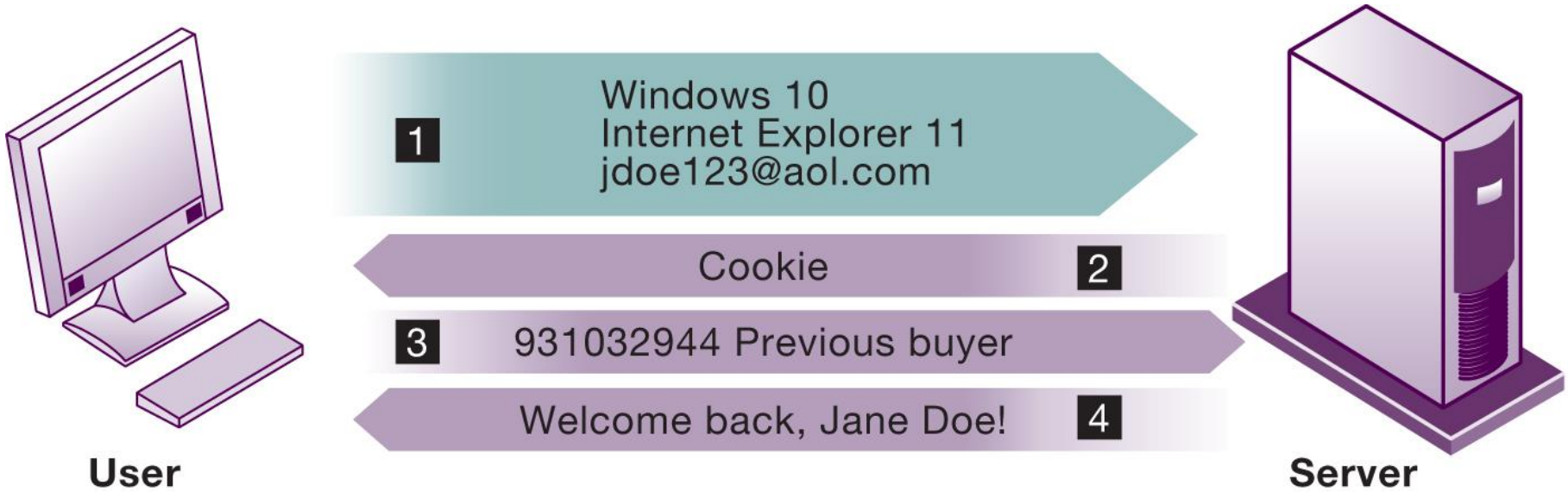
Internet Challenges to Privacy: Cookies

- **Websites and advertising networks routinely *track* users' searches, page visits, and purchases, often without users being fully aware of the extent of this monitoring.**
  - Tracking data are used to *target* advertisements and *customize* website content

- **Websites store small text files called *cookies* on users' devices to record browsing behavior and enable personalization**
  - Google Marketing Platform use cookies to analyze visitor behavior and online purchasing patterns.

- **Although cookies do not directly identify individuals, *combining cookie data with other information* can create detailed user profiles, raising significant privacy concerns.**

How Cookies Identify Web Visitors



1. The web server reads the user's web browser and determines the operating system, browser name, version number, Internet address, and other information.
2. The server transmits a tiny text file with user identification information called a cookie, which the user's browser receives and stores on the user's computer hard drive.
3. When the user returns to the website, the server requests the contents of any cookie it deposited previously in the user's computer.
4. The web server reads the cookie, identifies the visitor, and calls up data on the user.

Web Beacons & Spyware

- **Web Beacons (Web Bugs):**
  - Small software programs ***embedded in emails and web pages*** to monitor user behavior online.
  - Extent of Web Tracking:
    - Research found widespread surveillance, with tracking files discovered on most popular websites.
    - Major trackers include Google, Microsoft, and Quantcast, primarily selling user data to advertising firms.

- **Spyware:**
  - ***Can install itself*** on users' computers surreptitiously, reporting user movements online and delivering unsolicited ads.

Google's Dominance in Data Collection

- **Google is the largest collector of online user data globally, with about 80% of Internet users using Google services.**
  - Acquisitions like DoubleClick enabled ***behavioral targeting*** for more relevant ads based on user search activities.

- **Impact on Privacy:**
  - Google's extensive data collection capabilities raise concerns about ***online privacy***, as it possesses more personal information than many government agencies.
  - Ad programs like AdSense and Google Marketing Platform enable targeted ads across various platforms, including YouTube and mobile applications.

Current U.S. Privacy Practices and Challenges

- **Many U.S. businesses use customer transaction data without explicit consent, relying mainly on *opt-out models***
  - Microsoft, Yahoo, and Google publish their own privacy policies and offer opt-out options.

- **Opt-out tools are inconsistently implemented across browsers and websites, and most users do not actively use them.**

- **Although consumers express concern about privacy, companies often prioritize data-driven revenue, and *privacy policies remain complex, ambiguous, and rarely read*.**

Technical Solutions

- **Technological Solutions for Privacy:**
  - Tools like ***email encryption, anonymous browsing, cookie blockers, and spyware detection*** help protect user privacy during online interactions.
  - However, they often fail to prevent tracking as users move across websites.

- **Focus on Browsers:**
  - Due to public criticism and concerns about behavioral tracking and ad targeting, attention has shifted to ***web browsers***.
  - Many browsers offer ***Do Not Track*** options, ➔ However, websites are not obligated to honor these requests.
  - ***Private browser encryption*** software and mobile apps offer users a means to keep their online activities more private

Intellectual Property

- **Intellectual property includes tangible and intangible creations of the mind produced by individuals or organizations.**

- **Modern IS make it *easy to copy* and distribute digital content, putting pressure on traditional IP protections.**

- **Legal protections for IP rely on mechanisms such as *copyright*, *patents*, and *trade secrets*.**

- ***Copyright* protects the expression of a work for a limited time but does not protect underlying ideas, allowing others to recreate similar concepts without infringement.**

Other IP Protections

- **Patents: Grant exclusive monopoly rights to inventors for 20 years.**
  - Intended to ***reward inventors financially*** while promoting the widespread use of inventions.
- **Trademarks: Symbols used to distinguish products in the market.**
  - Infringement occurs when a firm uses another's marks without authorization or weakens their value.
- **Trade Secrets: Protect intellectual work products used for *business purposes.***
  - Protection through nondisclosure agreements is required to prevent them from entering the public domain.
  - The limitation lies in preventing ideas from becoming public when the software is widely distributed.

Piracy and Copyright in Networked Environments

- **Enforcing copyright is *difficult* because digital content can be easily copied, altered, and shared.**
  - Digital networks enable rapid, large-scale distribution, increasing piracy.
- **File-sharing platforms allowed users to share music and movies directly with each other, bypassing traditional distributors and copyright controls.**
  - Their decentralized structures made legal enforcement difficult, allowing piracy to continue even after shutdown efforts.
- **Legal measures such as the DMCA and legitimate platforms like iTunes, Kindle, and Spotify emerged to *reduce piracy and protect copyrights*.**
  - Digital Millennium Copyright Act (DMCA)

4.4

How have information systems affected laws for establishing accountability, liability, and the quality of everyday life?

Computer-Related *Liability* Problems

- **Traditional liability laws *face challenges* in assigning responsibility when software-driven systems or digital platforms cause harm.**
  - Major data breaches highlight unresolved questions about corporate liability for economic and personal damage.
- **Software producers are often difficult to hold liable, though some cases show the serious risks when software failures affect physical safety.**
- **SaaS systems and online platforms operate under *limited-liability frameworks* in the U.S., making accountability for defective or harmful software services unclear.**

System Quality, Reliability, and Software Limitations

- **Major cloud providers have experienced outages that disrupted widely used services. ➔ raise concerns about the *reliability and quality* of cloud-based systems.**
- **Deciding when a system is "good enough" to release involves a moral and economic trade-off.**
  - Perfect software is rarely feasible due to time and cost constraints.
- **Poor system performance commonly results from software bugs, hardware failures, and low-quality input data.**
- **The software industry faces ongoing responsibility for *system and data quality.***
  - A lack of standardized software testing and persistent data quality issues continue to affect performance.

Big Tech, Market Power, and Antitrust Challenges

- **Big Tech firms dominate major digital markets such as e-commerce, search, social media, streaming, and mobile platforms.**
  - Their economic power often translates into political influence through lobbying.
- **Antitrust regulation *has shifted* from promoting competition to focusing mainly on consumer prices and short-term welfare.**
  - Free or low-cost digital services complicate traditional antitrust analysis.
- **Growing public scrutiny has led to legislative investigations and calls for *antitrust reform*.**
  - Critics argue that Big Tech harms competition by limiting market access for innovators and threatening the survival of independent firms.
  - Some proposals include breaking up Big Tech to restore competition.

Rapidity of Change & Maintaining Boundaries

- **Rapidity of Change: Reduced Response Time to Competition**
  - IS increase efficiency and speed in global markets, but they also force businesses to ***respond to competition*** much faster than before.
  - The rise of a JIT society has led to ***on-demand work,*** such as Uber, creating flexible jobs but often without traditional employee benefits or insurance.

- **Maintaining Boundaries: Family, Work, and Leisure**
  - Ubiquitous computing and smartphones ***blur the boundaries*** between work and personal life, often extending work responsibilities into personal time.
  - ***Constant connectivity*** can strain family relationships and contribute to social problems such as cyberbullying and other antisocial behaviors.

Dependence and Vulnerability & Computer Crime and Abuse

- **Dependence and Vulnerability**
  - The critical reliance of businesses, governments, and other sectors on IS.
  - The potential consequences of system failures and the lack of regulatory or standard-setting forces.

- **Computer Crime and Abuse**
  - The increase in opportunities for computer crime such as ***malware, phishing, and denial of service attacks***.
  - Ethical concerns around ***computer abuse*** like spamming and its impact on businesses and individuals.
  - Regulatory Responses to Spamming

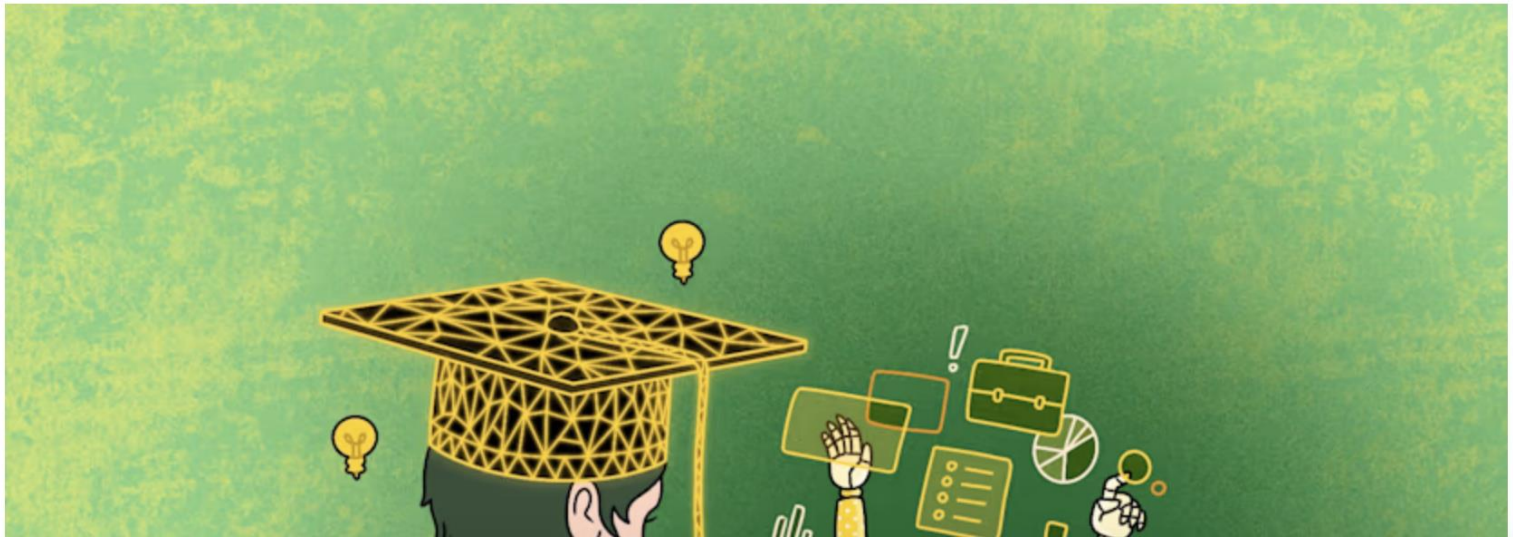
Employment & Equity and Access

- **Employment: Trickle-Down Technology and Reengineering Job Loss**
  - IS and automation can ***eliminate many mid-level and clerical jobs***, even as they create debate about whether technology destroys or creates jobs.
  - Adapting to these changes requires ***retraining*** the workforce, which is especially challenging for unskilled workers and less-educated managers.

- **Equity and Access: Increasing Racial and Social Class Divides**
  - The digital divide reflects ***unequal access*** to technology, which can widen social and economic ***gaps*** and affect how different groups use the internet.
  - Efforts to bridge this divide focus on expanding access to digital resources, similar to past initiatives that made telephone service widely available.

AI is not wiping out all entry-level jobs, but it's changing the game and fresh jobseekers need to level up

Experts say AI isn't wiping out all junior roles, but it's forcing fresh grads to level up and prove the one thing machines still can't replace: human judgment.



Amanda Yea

07 Nov 2025 09:30PM
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Health risks: RSI, CVS, and Cognitive Decline

- **Health Risks Associated with IT**
  - Repetitive stress injuries (RSI), especially carpal tunnel syndrome (CTS).
  - The impact of poorly designed workstations on back, neck, leg, and foot pain.
- **Ergonomic solutions to prevent RSI, and emphasis on regular breaks and job rotation**
  - e.g., adjustable furniture, wrist rests, and proper screen placement.
- **Computer Vision Syndrome (CVS)**
- **Cognitive Considerations**
  - Concerns about the effect of ***persistent computer use*** on cognitive functions, such as focus and problem-solving.
  - The debate over whether technology hinders learning and independent thinking.

Video

- **A data broker defends his industry (02:33)**

Discussion

- **Some people may like personalized ads because they can be more useful and relevant. But if it feels like the phone is listening to private conversations, many people would find that uncomfortable. So the real question is whether convenience is worth the loss of privacy.**

  Are personalized ads worth the privacy risk if your phone seems to be listening to you? Why or why not?

Key Takeaways

- Rapid advances in IT are outpacing existing laws and social norms, creating new ethical, legal, and political challenges around privacy, intellectual property, accountability, and quality of life.
- Ethical decision making in IS can be guided by a structured five-step analysis process and supported by ethical principles
- Modern data collection and analytics enable extensive tracking and profiling of individuals, often without strong privacy protections or informed consent, while digital copying weakens traditional copyright enforcement.
- Although IS increase efficiency and wealth, they also introduce risks such as system errors, job displacement, inequality, computer crime, and negative health and cognitive effects.

Next Week...

- **Next week, we will cover:**
  - [Chapter 7] Telecommunications, the internet, and wireless technology*

Q & A